Funcion de transferencia

Hipotenso

$$\frac{P_p(s)}{P_a(s)} = \frac{LR_s + R_2}{CLR_2s^2 + L(2+R)s + R_2}$$

$$= \frac{1\times10^{-3}s + 0.01}{3.5\times10^{-5}s^2 + 1.01\times10^{-3}s + 0.01}$$

Normotenso

$$\frac{P_p(s)}{P_a(s)} = \frac{1\times10^{-3}s + 0.01}{1.5\times10^{-5}s^2 + 1.01\times10^{-3}s + 0.01}$$

Hipertenso

$$\frac{P_p(s)}{P_a(s)} = \frac{1\times10^{-3}s + 0.01}{1.5\times10^{-5}s^2 + 1.01\times10^{-3}s + 0.01}$$

~~Normotenso~~ Hipotenso

$Z = 0.01$, $L = 0.001$, $R = 1.000$, $C = 3.500$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4(CLR^2Z^2}}{2CLRZ}$$

$CLRZs^2 = a + L(Z+R)s\ b\ ,\ RZ = 0$

$\lambda_1 = -14.4286 + 8.8052i$

$\lambda_2 = -14.4286 + 8.8052i$

* El sistema presenta una respuesta estable y subamortiguada a un escalón unitario

Normotenso

$Z = 0,01$, $L = 0,001$, $R = 1,00$, $C = 1.500$

$\lambda_1 = -12.0616$

$\lambda_2 = -55.2717$

El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario

Hipertenso

$Z = 0,010$, $L = 0;001$, $R = 1,000$, $C = 1.000$

$\lambda_1 = -11.2 68$

$\lambda_2 = -89.8732$

El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario

$$Pa(s)\frac{Ls+Z}{LZs}=Pp(s)\frac{CLRZs^2+L(Z+R)s+RZ}{RLZs}$$

$$\frac{Pp(s)}{Pa(s)}=\frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2+L(Z+R)s+RZ}{RLZs}}$$

Resultado

$$\frac{Pp(s)}{Pa(s)}=\frac{LRs+RZ}{CLRZs^2+L(Z+R)s+RZ}$$

Estabilidad del sistema en lazo abierto

Hipotenso

$$CLRZs^2+L(Z+R)s+RZ=0$$

Z = 0.020 | $a=CLRZ$

C = 0.250 | $b=LC(Z+R)$

R = 0.600 | $C=RZ$

L = 0.005 |

$$\lambda_1\lambda_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-L(R+Z)\pm\sqrt{L^2(Z+R)^2-4CLR^2Z^2}}{2a=2CLRZ}$$

ontario de entrada.

Error en estado estacionario.

$$e(s) = \lim_{t \to 0} sPa(s)\left(1 - \frac{Pp(s)}{Pq(s)}\right) =$$

Se calcula a un escalón unitario de entrada:

$$= e(s) = \lim_{t \to 0} \frac{1}{s}\left(1 - \frac{LRs + RZ}{CLRZ\frac{2}{s} + L(Z+R)s + RZ}\right) = 0$$

$$e(t) = 0V$$

Modelo de ecuaciones integro - diferenciales

$$Fz(t) + FL(t) = Fc(t) + FR(t)$$

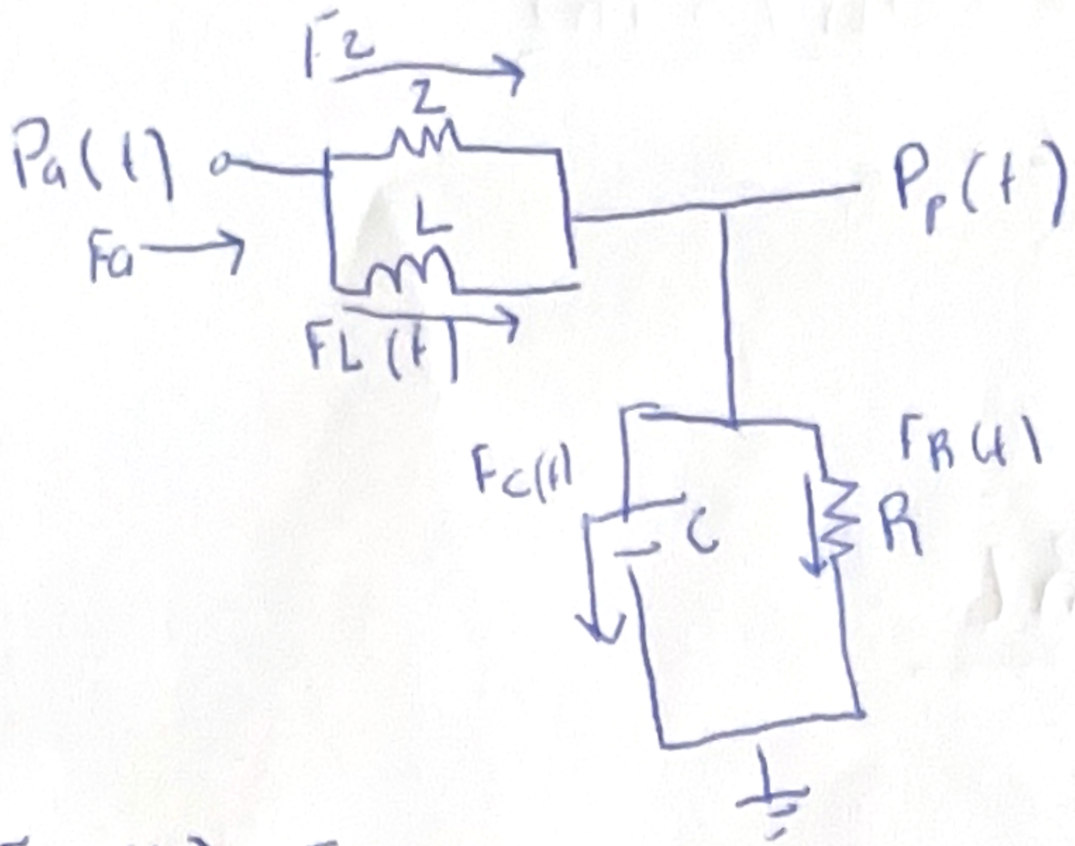
Nota: No se pueden dejar derivadas e integrales

$$\frac{Pa(t)}{Z} + \frac{1}{L}\int |Pa(t) - Pp(t)|\,dt - C\frac{dPp(t)}{dt} =$$

$$= \frac{Pp(t)}{R} + \frac{Pp(t)}{Z}$$

$$\left(\frac{1}{R}\right) + \left(\frac{1}{Z}\right)) Pp(t) = \frac{Pa(t)}{Z} + \frac{1}{L}\int |Pa(t) - Pp(t)|\,dt - C\frac{dPp(t)}{dt}$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int |Pa(t) - Pp(t)|\,dt - C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$



$$Fa(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{Pa(t) - Pp(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int Pa(t) - Pp(t)\, dt$$

$$F_C(t) = C\frac{dPp(t)}{dt}$$

$$F_R(t) = \frac{Pp(t)}{R}$$

Transformada de Laplace

$$Fa(s) = F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_Z(s) = \frac{Pa(s) - Pp(s)}{Z}$$

$$F_L(s) = \frac{Pa(s) - Pp(s)}{Ls}$$

$$F_C(s) = CsPp(s)$$

$$F_R(s) = \frac{Pp(s)}{R}$$

Procedimiento algebraico

$$F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$\frac{Pa(s) - Pp(s)}{Z} + \frac{Pa(s) - Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

$$\frac{Pa(s)}{Z} - \frac{Pp(s)}{Z} + \frac{Pa(s)}{Ls} - \frac{Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

$$Pa(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = Pp(s)\left(s + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$$

$$Pa(s)\frac{Ls + Z}{LZs} = Pp(s)\left(\frac{CLRZs^2 + LZs + RZ + RL}{RLZs}\right)$$